



TASHKENT UNIVERSITY OF TECHNOLOGY

“APPROVED”
Vice-Rector for Academic Affairs
A.Rustamov

August “___”, 2026

“LINEAR ALGEBRA – I”
Course Syllabus

Field of knowledge: 700000 – Engineering, processing and construction industries

Field of study: 710000 – Engineering work

Course of Study: 60710400 – Energy engineering

60710500 – Electric engineering

60712200 – Industry engineering and management

60720300 – Materials Science

Tashkent – 2026

This curriculum has been approved by the Resolution No. ____ of the meeting of the University Council dated ____ August 2026.

Prime Instructor:

Dr. Giorgos Tsironis – Professor, Tashkent University of Technology.

External Reviewer:

Dr. G. D. Barmparis, Department of Physics, University of Crete, Greece

Reviewed on Faculty of Energy engineering meeting
August «__» 2026, act № ____

Dean of Faculty of Energy engineering

M.R. Bahrami

Approved

Head of Academic affairs department

Head of Education Quality Control
department

Head of Information Resource Centre

“LINEAR ALGEBRA – I”

Course Syllabus

1. Subject/Module Information

Field	Content
Subject/Module Name	Linear Algebra – I
Subject/Module Code	CC-LIAL010102
Subject/Module ECTS	6
Academic year	2026 - 2027
Study Semester	Fall (semester 1)
Subject/module type	Mandatory
Program Affiliation	Bachelor of Science for all tracks
Language of Instruction	English
Prime Instructor	Dr. Giorgos Tsironis

2. Forms of Learning and Teaching

Workload breakdown (30 hrs/ECTS convention):

Component		Hours
Total Classroom Hours		90
	Lecture	60
	Practical classes (problem-solving and group work)	30
Independent Study		90
Total Workload		180

3. Module Description

This course covers matrix theory and linear algebra, with emphasis on topics useful in other disciplines. It develops methods for solving systems of linear equations and understanding the properties and representations of matrices and vector spaces. The course includes linear regression, Fourier methods, eigenvalues and eigenvectors, matrix diagonalization, symmetric and positive definite matrices, and singular value decomposition. Linear algebra concepts are applied in contexts including physics, data science, and robotics.

4. Entry Requirements

Field	Content
Prerequisites	No
Co-requisites	No
Additional Remarks	No

5. Intended Learning Outcomes (ILOs)

No	Bloom Level	ILO	ASIIN Dimension
1	Understand	Explain fundamental concepts of vector spaces, subspaces, linear independence, basis, and dimension; represent and solve systems of linear equations using matrix methods (Gaussian/Gauss–Jordan elimination, rank, inverses, LU/LDU factorization).	Subject-specific
2	Analyze	Analyze eigenvalues, eigenvectors, and diagonalization; evaluate symmetric, positive definite, similar, and singular-value-decomposition representations of matrices and quadratic forms; analyze complex, Hermitian, and unitary matrices and apply Fourier-transform concepts.	Subject-specific
3	Select	Select the most effective method — elimination, LU/LDU factorization, Gram–Schmidt orthogonalization, QR decomposition, or eigen-decomposition — for a given linear-algebra or regression problem, and justify the choice of approach based on the structure of the problem.	Methodological
4	Evaluate	Judge which linear-algebra representation or technique (diagonalization, eigen-decomposition, singular value decomposition) is appropriate for a given engineering or data-driven application (e.g. physics, data science, robotics), evaluating its role within that broader context.	Systemic
5	Apply	Collaborate in small groups on assigned projects and clearly communicate mathematical solution strategies and results, in written and verbal form, to peers.	Social

6. Course Content (Weekly Breakdown)

Week	Topic	Format	Hours
1	Introduction; vectors; definition and operations; vector addition; scalar-vector multiplication; properties of vector matrices; vector space axioms; span; linear independence	Lecture	4
		Practice	2
2	Dot product; cross product; introduction to matrices; operations with matrices	Lecture	4
		Practice	2
3	Matrices: transpose, addition, scalar multiplication, matrix multiplication; determinants; scalar triple product	Lecture	4
		Practice	2
4	Matrix rank; matrix inverse + Test 1	Lecture	4
		Practice / Assessment	2
5	Geometry of linear equations; elimination with matrices	Lecture	4
		Practice	2
6	Matrix operations, including inverses; LU and LDU factorization	Lecture	4
		Practice	2
7	Transposes and permutations; vector spaces and subspaces	Lecture	4
		Practice	2
8	Midterm	Assessment	6

Week	Topic	Format	Hours
9	The null space: $Ax = 0$; row-reduced echelon form; fundamental subspaces; matrix rank	Lecture	4
		Practice	2
10	Independence, basis and dimension; the four fundamental subspaces	Lecture	4
		Practice	2
11	Orthogonal vectors and subspaces; projections onto subspaces	Lecture	4
		Practice	2
12	Projection matrices; least-squares approximations; Gram–Schmidt and $A = QR$	Lecture	4
		Practice	2
13	Test 2	Assessment	6
14	Eigenvalues and eigenvectors; matrix diagonalization	Lecture	4
		Practice	2
15	Symmetric and positive definite matrices; similar matrices; singular value decomposition (SVD); pseudoinverse	Lecture	4
		Practice	2
Total Hours			90

Assessment weeks are included in the 90 classroom hours. The lecture and practical workload totals above follow the course allocation; examination durations are specified in Section 9.

7. Independent Study Assignments*

Week	Assignments	Hours
1	Individual problem set: vector operations, vector-space axioms, span and linear independence exercises.	6
2	Individual problem set dot/cross product applications and introductory matrix-operation exercises.	6
3	Small-group problem set: matrix algebra (transpose, scalar multiplication, matrix multiplication), determinants and the scalar triple product.	6
4	Individual problem set: rank determination and matrix-inverse computation via Gauss–Jordan elimination, with worked case examples.	6
5	Small-group problem set: geometric interpretation of linear systems and elimination with matrices.	6
6	Individual problem set: LU and LDU factorization exercises.	6
7	Individual problem set: verifying vector-space and subspace properties (transposes, permutations, subspace criteria).	6
8	Individual self-assessment: consolidated review and practice problems in preparation for the Midterm.	6
9	Individual problem set: null space ($Ax = 0$), row-reduced echelon form, and the fundamental subspaces.	6
10	Small-group problem set: independence, basis, dimension, and the four fundamental subspaces.	6
11	Small-group problem set: orthogonal vectors/subspaces and projection exercises applied to a shared dataset or example system.	6
12	Individual problem set: projection matrices, least-squares approximation, Gram–Schmidt orthogonalization, and QR decomposition.	6
13	Individual self-assessment: consolidated review and practice problems in preparation for Test 2	6
14	Individual problem set: eigenvalues, eigenvectors, and matrix diagonalization.	6

Week	Assignments	Hours
15	Small-group problem set: symmetric/positive-definite matrices, singular value decomposition (SVD) and pseudoinverse case study, with course wrap-up review.	6
Total Hours		90

***Note: Assignments in weeks 3, 5, 10, 11 and 15 require collaborative learning in small groups (five of fifteen assignments). Students should communicate their solution strategies and results clearly. The remaining assignments are individual tasks.**

8. Teaching and Learning Methods

- Lectures
- Seminars / laboratory classes
- Homework and group projects
- Discussions
- Testing (written or computer based)

9. Assessment Methods and Completion

Component	Examination Type	Duration/Length	Weight (%)	ILOs
Midterm Exam	Paper-based problem solving	1–2 hours	30	1, 3
Test 1	Paper-based problem solving	1–2 hours	15	1, 3
Test 2	Paper-based problem solving	1–2 hours	15	2
Final Exam	Paper-based problem solving	1–2 hours	40	1–4

Grading and Completion

90-100 points = A - Excellent

75-89 Points = B - Good

60-74 Points = C - Satisfactory

0 - 59 Points = D - Poor

Student Knowledge Assessment Criteria

“A” mark is put to the student who can freely operate covered materials; does not make mistakes; actively participates in the process of communication; gives full and detailed answers.

“B”, mark is put to the student who knows the material well, and can express it in a clear and logical way; actively participates in the process of communication; formulates full and detailed answers, but has minor inaccuracies and mistakes.

“C” mark is put to the student who has knowledge of basic material but has not obtained details, has inaccuracies; gives not enough correct formulations while answering; breaks logical correction in presenting material; faces difficulties in the process of communication.

“D” (FAIL) mark is put to the student who does not have an idea on the essence of the question; does not have answers; does not participate in the process of communication.

Students who miss more than 25% of the total class hours without a valid excuse are not permitted to take the final assessment.

While assessing students' works the following is taken into consideration:

- attendance;
- participation;
- study of core and additional materials;
- timely submission of self-study tasks;
- timely submission of all tasks.

10. Indicative Literature

- G. Strang, Introduction to Linear Algebra, Sixth Edition, Wellesley–Cambridge Press, 2023. ISBN: 978-17331466-7-8.
- Gilbert Strang, Linear Algebra and Its Applications, 4th Edition, Brooks Cole, 2006. ISBN: 9780030105678.

11. Statement on Plagiarism and Academic Dishonesty:

Tashkent University of Technology observes a zero-tolerance policy regarding academic dishonesty. Per university policy as described in the academic catalog, all cases of academic dishonesty will be reported. The following policy will be used for this course:

1. First instances of academic dishonesty will result in a 0 grade on the assignment, but a second attempt may be allowed at the instructor's discretion.
2. Second instances of academic dishonesty will result in a 0 grade on the assignment with no opportunity to reattempt.
3. Third instances of academic dishonesty will result in automatic failure of the course.

12. Statement on the use of AI-tools

Tashkent Tech recognizes artificial intelligence (AI) as a powerful tool for learning, innovation, and skill development. Students are encouraged to use generative AI tools (such as ChatGPT, Claude, DeepSeek or similar platforms) to clarify concepts, debug code, and improve their learning through dialogue (with the AI tools). When used responsibly, AI can serve as a valuable collaborative partner. However, any content generated by AI must be treated as a source, meaning that students must explicitly cite the tool used and clearly document how it assisted you in their workflow, in accordance with each course's specific guidelines.

Academic integrity remains the fundamental principle governing AI use. Students may not submit AI-generated text, code, calculations, or other outputs as their own work unless explicitly authorized by the instructor. All graded assignments, essays, and exams must reflect the student's own understanding and be explainable in detail upon request. The unauthorized or unacknowledged use of AI to complete assignments will result in academic penalties, as stated in the *Statement on Plagiarism and Academic Dishonesty* guidelines.